

GUSTAVO CARVALHO

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EDUCATION

University of São Paulo – USP *Present*
PhD Candidate in Electrical Engineering, *Power Systems Economics*

Pontifical Catholic University of Rio de Janeiro – PUC-Rio
MSc in Electrical Engineering, *Operations Research* *2025*
Extension Program in Risk Analysis *2011*
BSc in Electrical Engineering, *Decision Support Systems* *2011*

Getulio Vargas Foundation – FGV *2017*
Executive MBA in Finance

PROFESSIONAL EXPERIENCE

AXIA Energia **São Paulo, SP**
Market Strategy Director *September 2025 – Present*

Responsible for leading AXIA Energia’s integrated commercialization and market development agenda, encompassing Market Analytics, Structured Energy and Derivatives Products, Environmental Assets, New Ventures, Strategic Partnerships, and Commercial Marketing. Drives the company’s revenue maximization strategy by transforming structural market shifts, including the energy transition, increasing volatility, digitalization, and evolving customer demands, into scalable and profitable business opportunities. Leads the development of innovative solutions integrating physical energy, derivatives, flexibility, and environmental attributes, while advancing AXIA Energia’s positioning in Brazil and internationally through strategic partnerships, market innovation, and new business models. Recipient of the “Vicente Falconi Award 2025” for outstanding performance and business results.

AXIA Energia **São Paulo, SP**
Market Strategy Executive Manager *Jan 2024 – August 2025*

Led AXIA Energia’s market strategy and commercial planning initiatives. Played a central role in embedding data-driven decision-making into the company’s commercial and portfolio management processes, while designing innovative commercial models that expanded competitiveness and strengthened market positioning.

Oliver Wyman **São Paulo, SP**
Engagement Manager (Energy and Natural Resources) *Oct 2022 – Jan 2024*

Contributed to the establishment and expansion of Oliver Wyman’s Energy and Natural Resources practice in Brazil, leveraging the firm’s global expertise to support strategic projects in the Brazilian energy sector. Led consulting engagements focused on market dynamics, commercial strategy, and sector positioning, including advisory work for multinational energy companies seeking to strengthen their competitive positioning and understand liquidity dynamics in the Brazilian power market.

Thymos Energia
Pricing and Market Research Manager

São Paulo, SP
Jun 2020 – Oct 2022

Led the Pricing and Market Research area, specializing in PLD projections, market intelligence, and strategic studies for the Brazilian electricity sector. Supported major economic groups in strategic planning and risk management initiatives, contributed to the CCEE technical security note market report, and advised both trading companies and international investors on middle office structures, risk policies, and market entry strategies in Brazil.

Brookfield Energia Renovável
Portfolio Manager

Rio de Janeiro, RJ
Sep 2018 – Feb 2020

Led the restructuring of Brookfield's portfolio management area, overseeing generation portfolio optimization and energy hedging strategies across the company's assets. Implemented initiatives that significantly improved operational efficiency and generated market-related returns exceeding R\$60 million annually.

Enel Brasil
Portfolio Management Sr. Specialist

Niterói, RJ
Apr 2018 – Aug 2018

Supported the management and optimization of Enel's diversified generation portfolio, including hydro, wind, solar, and thermal assets. Managed short- and long-term contracts in the Brazilian Free Contracting Environment (ACL), contributing to portfolio balancing and hedging strategies.

Neoenergia
Generation Regulatory Affairs Specialist

Rio de Janeiro, RJ
May 2016 – Apr 2018

Worked on regulatory affairs and strategic regulatory analysis for key generation assets within the Neoenergia Group, including hydroelectric and thermal projects. Acted in coordination with ANEEL and industry associations such as ABRAGET and APINE to support the company's regulatory positioning and sector representation.

CPFL Energia
Energy Planner

Campinas, SP
Oct 2014 – May 2016

Responsible for energy market forecasting and strategic planning activities for the group's distribution companies, preparing analyses and executive presentations to support senior management decision-making processes.

Engenho Pesquisa e Consultoria
Researcher

Rio de Janeiro, RJ
Jul 2014 – Sep 2014

Conducted research and monitoring activities focused on the Brazilian electricity sector, with emphasis on energy consumption forecasting for distribution companies.

Siglasul Consultoria
Distribution Regulatory Affairs Consultant

Rio de Janeiro, RJ
May 2012 – Jul 2014

Worked on tariff review and adjustment projects for companies in the electricity and sanitation sectors, including AES Sul, Afluente T, and Sanepar. Activities involved regulatory and financial modeling, market analysis, tariff structures, investments, losses, WACC calculations, and scenario evaluation.

Núcleo de Pesquisa em Energia e Infraestrutura da PUC
Research Assistant

Rio de Janeiro, RJ
Jan 2011 – Apr 2012

Conducted research and consulting activities focused on regulation, finance, and business strategy for the Brazilian electricity sector, supporting studies to infrastructure and energy markets.

Light ESCO
Energy Trading Intern

Rio de Janeiro, RJ
Mar 2009 – Mar 2010

Supported energy trading and portfolio management activities using optimization models such as NEWAVE and DECOMP to simulate spot price (PLD) scenarios and manage energy contract portfolios.

PROFESSIONAL AFFILIATIONS

CIGRE – Conseil International des Grands Réseaux Électriques
Member, International Working Group on Retail Electricity Market Competition

IEEE – Institute of Electrical and Electronics Engineers
Member, Power & Energy Society Long Range Planning Subcommittee

HONORS AND AWARDS

Vicente Falconi Award	AXIA Energia	2025
IEEE Senior Member Elevation	IEEE	2025
MSc Scholarship	CAPES	2023
Top-Ranked Student in Finance, all cohorts	FGV	2017
Undergraduate Research Grant	CNPq	2006
BSc Scholarship	PUC-Rio	2006

PUBLISHED PAPERS

1. Alexandre Street, Bruno Fanzeres, Gustavo Carvalho, **Swaps Horários Baseados em Complementariedade Renovável no Brasil: Uma Abordagem de Equilíbrio de Mercado para Mitigação de Riscos**, *XXVIII Seminário Nacional de Produção e Transmissão de Energia Elétrica*, October 2025 
2. Alexandre Street, Bruno Fanzeres, Gustavo Carvalho, **Managing Renewable Energy Risk in the Brazilian Forward Market: The Impact of Hourly Hedging**, *IEEE Transactions on Energy Markets, Policy and Regulation, Volume: 3, Issue: 1*, March 2025 
3. Victor Ribeiro, João Carlos Mello, Alexandre Guedes Viana, Gustavo Carvalho, Lucas Cechetto, Vinicius David, Leonardo Calabro, Pedro Moro, **Implementação do Mecanismo de Preço por Oferta no Sistema Interligado Nacional (SIN) - Uma Discussão Estruturada**, *XXVI Seminário Nacional de Produção e Transmissão de Energia Elétrica*, May 2022 
4. Alexandre Street, Gustavo Carvalho, Bruno Fânzeres dos Santos, Davi Michel Valladão, **Desenho e Precificação de Contratos em Base Horária para Hedge Intradiário de Fontes Renováveis**, *XXVI Seminário Nacional de Produção e Transmissão de Energia Elétrica*, May 2022 

5. Gustavo Carvalho, Alexandre Guedes Viana, João Carlos Mello, **Os Contratos Derivativos e o Setor Elétrico – Uma Solução Legítima para a Segurança do Mercado Nacional**, *XXVI Seminário Nacional de Produção e Transmissão de Energia Elétrica*, May 2022 [!\[\]\(effbd7993c63c039a58fd3395789cf3f_img.jpg\)](#)
6. Gustavo Carvalho, **Como o setor elétrico pode se beneficiar com o uso dos contratos derivativos?**, *Thymos Research*, April 2021 [!\[\]\(144980d038f2541d7b588a8a9132bd70_img.jpg\)](#)
7. Gustavo Carvalho, **A questão do GSF e a lição de Edmund Burke**, *Canal Energia*, September 2020 [!\[\]\(c4ce2d477989700c971cf3d240ad9283_img.jpg\)](#)
8. Hugo Nunes, Gustavo Carvalho, Raphael Carvalho, Laura Gunn, Pablo Ribeiro, **Interdependência Entre a Repactuação do Risco Hidrológico e a Aversão ao Risco Utilizada nos Modelos de Formação de Preços: Sugestão de Aperfeiçoamento Regulatório**, *XXIV Seminário Nacional de Produção e Transmissão de Energia Elétrica*, October 2017 [!\[\]\(5013555a72072875cb154b597e002a46_img.jpg\)](#)

SELECTION OF PRESS INTERVIEWS

1. Para especialistas, horário de verão ajuda a aliviar pressão..., **Agência Estado**, 09/2022 [!\[\]\(13b6bdd0ca077c333d50231f1443cb1d_img.jpg\)](#)
2. Bandeiras antecipam custos, mas todos precisariam pagar..., **Valor Econômico**, 07/2022 [!\[\]\(5dbedd4e1e8871e3a0e67053ad2f9701_img.jpg\)](#)
3. Por que o futuro do mercado de energia passa pelos derivativos?, **Head Energia**, 06/2022 [!\[\]\(d4749465acb9b53e115af1f9ce82539c_img.jpg\)](#)
4. Alívio na conta de luz sem sobretaxa é menor que o prometido..., **O Globo**, 04/2022 [!\[\]\(3e3001313d495ec87b5a6a5de6205728_img.jpg\)](#)
5. Bolsonaro anuncia fim de taxa extra na conta de luz..., **Estadão**, 04/2022 [!\[\]\(e26df985e6d3e053d2593dc7b93b41cf_img.jpg\)](#)
6. Energia cara puxa investimentos..., **Valor Econômico**, 02/2022 [!\[\]\(2d8989e35a5d1c61f2b9b0307dee0da4_img.jpg\)](#)
7. Com risco de apagão, empresas investem R\$bi em termelétricas..., **O Globo**, 09/2021 [!\[\]\(10225a66c9f99322b84a7fc32767a3b8_img.jpg\)](#)
8. Prevalência do clima sobre incentivos para redução de consumo, **Brasil Energia**, 09/2021 [!\[\]\(e628152647eaf2a5b8bfa082bbb75081_img.jpg\)](#)
9. Brazilian power companies' working capital needs increase..., **Debtwire**, 09/2021 [!\[\]\(322137fe39b9b64ec6280cbae636a504_img.jpg\)](#)
10. Bolsonaro distorce dados, e hidrelétricas não devem ter colapso..., **Folha de São Paulo**, 09/2021 [!\[\]\(c7b4c0f4f14640e9a5f8748e3c49f893_img.jpg\)](#)
11. Sistema no limite eleva risco de apagão..., **O Globo**, 08/2021 [!\[\]\(983d55426a5421081cd9eaf8830e7192_img.jpg\)](#)
12. Especial: 2022 é logo ali, **Canal Energia**, 07/2021 [!\[\]\(0265fc2a1f3b606d78de384d8ebca901_img.jpg\)](#)
13. Com reajuste da bandeira tarifária..., **Broadcast Energia Estadão**, 06/2021 [!\[\]\(60f08d4be84d697251258af68db7f264_img.jpg\)](#)
14. Especial: Tensão e Atenção no Período Seco, **Canal Energia**, 06/2021 [!\[\]\(e0a4f36696eaf68f18fcba0321218c0e_img.jpg\)](#)
15. Especialistas veem conta de luz mais cara..., **O Estado de SP**, 05/2021 [!\[\]\(4d15393fc2c10be98a56a95b7e2ea3f3_img.jpg\)](#)
16. Crise hídrica deve manter bandeira vermelha até 2022, **Canal Energia**, 05/2021 [!\[\]\(e15dabadb063dccee91c42caa8ad0dc7_img.jpg\)](#)
17. Conta de luz: Por que a tarifa pode ficar mais cara..., **BBC Brasil**, 04/2021 [!\[\]\(727df61877525c9a4d67d81103a57746_img.jpg\)](#)
18. Como os derivativos de energia podem contribuir..., **Live Thymos Research**, 04/2021 [!\[\]\(51d88d34502ab9fae984a661b011ed50_img.jpg\)](#)

- 19. Bandeira vermelha pode passar a valer já em maio..., **Agência Estado**, 04/2021 
- 20. O fantasma do racionamento, **Brasil Energia**, 04/2021 
- 21. Acordo em Belo Monte tem provisão de R\$150milhões, **Valor Econômico**, 02/2021 
- 22. Vazão menor em Belo Monte pode elevar preço da energia, **Valor Econômico**, 01/2021 
- 23. Hidrelétricas poderão perder R\$1bi com preço horário, **Energia Hoje**, 12/2020 